4. PROJECT PLANNING

EduGenie — Google Gemini Powered Learning Assistant

Development phases, milestones, deliverables, risks, schedule, and verification plan

Implementation Phases

Phase	EduGenie implementation
Phase 1 — Foundation	Create the FastAPI application, configure dependencies, establish the project folder structure, and prepare the
Phase 2 — Requirements & Validation	Define the supported learning tasks and input behavior for Q&A, explanation, quiz, summarization, and learning
Phase 3 — Core AI Generation	Connect Gemini 1.5 Pro for Q&A, quiz generation, summarization, and learning recommendations; implement t
Phase 4 — Learning Modules & Integration	Integrate the five modules with the FastAPI routes and standardize the returned responses for the frontend.
Phase 5 — Frontend & API	Build the Jinja2/HTML interface, CSS styling, task selector, input form, result display, and REST API endpoints.
Phase 6 — Testing & Documentation	Test each learning function, verify response handling, document installation and execution, and prepare the fina

Milestone Checklist

Milestone	Completion target	Verification
M1	FastAPI project starts and home page renders.	GET / or home page responds successfully.
M2	Learning task selection and input handling work.	Valid task/input is accepted; invalid or empty input is handled.
M3	AI generation modules produce results.	Q&A, explanation, quiz, summary, and learning-path outputs are c
M4	Quiz output is structured correctly.	Three MCQs, four options, and correct answers are returned in th
M5	Frontend and API integration work.	Form submission reaches the correct endpoint and displays the re
M6	Testing and documentation are complete.	Functional tests pass and setup/run instructions are documented.
M7	Final demonstration package is ready.	Application can be run locally and the main learning features can

Project Planning Flow

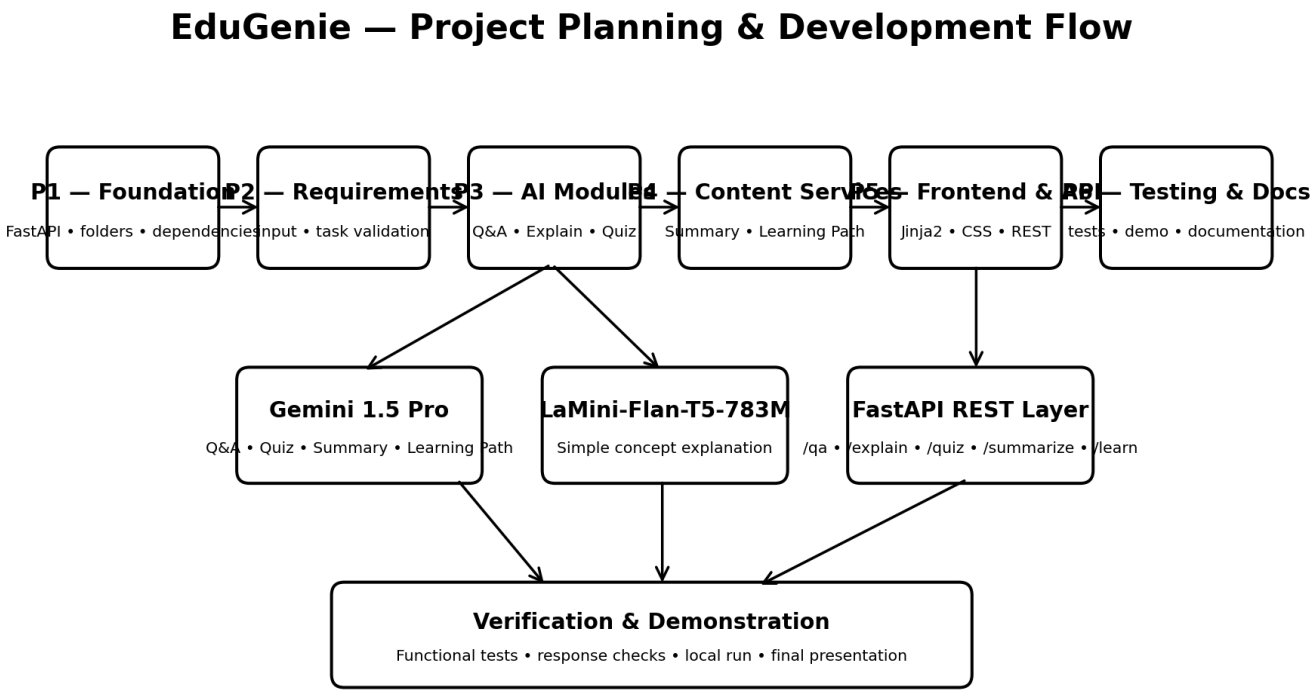


Fig. EduGenie development and verification flow

Risk and Mitigation

Risk	Mitigation
Gemini service unavailable	Use controlled local development/testing behavior and clear error handling; the project architecture keeps AI
Invalid or empty user input	Validate the selected task and text input before calling the AI service and display a useful message.
AI response format differs	Clean generated output before parsing; especially validate the quiz JSON structure before displaying it.
Model/API configuration changes	Keep API configuration and model selection in environment/configuration rather than hard-coding secrets.
Slow AI response	Keep prompts focused, use lightweight explanation processing where appropriate, and provide clear result/e
Testing differences between local and cloud AI	Use repeatable sample inputs and test endpoint behavior separately from live model quality.
Documentation/demo issues	Maintain a clear README/setup procedure and verify the application from a clean local run before presentat

Suggested Project Schedule

Following the reference planning format, the EduGenie project can be organized into eight academic working sessions: ideation, requirements, design, planning, development, testing, documentation, and final demonstration. Each session should finish with a concrete artifact that can be reviewed before moving to the next stage.

Session	Activity	Expected artifact
1	Ideation & problem definition	Project objective and feature list
2	Requirements analysis	Functional/non-functional requirement list
3	System design	Architecture and module design
4	Project planning	Phases, milestones, risks, and schedule
5	Development	FastAPI, AI modules, frontend, and API integration
6	Testing	Endpoint and feature verification results
7	Documentation	Project report, setup instructions, and screenshots/diagrams
8	Final demonstration	Working application and presentation flow

Quick Reference

Milestone	Primary deliverable	Verification
M1	Running application skeleton	Home page loads
M2	Validated learning input	Input/task behavior verified
M3	AI learning modules	Expected content generated
M4	Integrated module responses	Correct endpoint/module response
M5	Frontend/API layer	User can submit and view results
M6	Test and documentation package	Tests and setup instructions complete
M7	Demo package	Main EduGenie features demonstrated

Project basis: This planning document adapts the supplied Project Planning reference structure to the EduGenie learning-assistant project and preserves the reference's phases, milestone, risk, schedule, and verification style.